

Total No. of Questions : 8]

SEAT No. :

P7839

[Total No. of Pages : 3

[6181]-232

B.E. (Mechanical)

DYNAMICS OF MACHINERY

(2019 Pattern) (Semester-VII) (402042)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1) a) Determine the natural frequency of the mass 'm' placed at one end of a cantilever beam of negligible mass as shown in fig. 1a. [5]



Fig. 1a

- b) Explain with neat diagram mathematical model of a motorbike. [6]
- c) A circular cylinder of mass 4 kg and radius 15 cm is connected by a spring of stiffness 4000N/m as shown in Fig. 1c. If it is free to roll on surface without slipping, determine the natural frequency. [6]

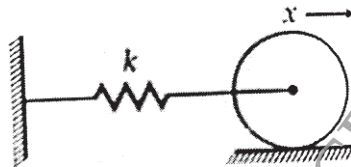


Fig. 1c

OR

P.T.O.

- Q2) a)** Define the term damping factor and explain with neat sketches, underdamped, overdamped and critically damped systems. [8]
- b)** An under-damped shock absorber is to be designed for a motorcycle of mass 200 kg such that during a road bump, the damped period of vibration is limited to 2 second and amplitude of vibration should reduce to one-sixteenth in one cycle. Find the spring stiffness and the damping coefficient of shock absorber. [9]

- Q3) a)** Derive an expression for the amplitude of vibration for a system having rotating or reciprocating unbalance. [8]
- b)** A machine part of mass 4 kg vibrates in a viscous fluid. Determine the damping coefficient when a harmonic exciting force of 50 N results in resonant amplitude of 250 mm with a period of 0.4 second. If the excitation frequency is 2 Hz, find the percentage increase in the amplitude of forced vibration when damper is removed. [10]

OR

- Q4) a)** A radio set of 20 kg mass has to be isolated from a machine vibrating with an amplitude of 0.05 mm at a speed of 500 cpm. The set is mounted on four isolators, each having a spring scale of 31400N/m and damping coefficient of 392 N-s/m. Determine [10]
- Amplitude of vibration of radio set
 - Dynamic load on each isolator due to vibration.
- b)** Derive an expression for deflection of vertical shaft with a single rotor without damping. [8]
- Q5) a)** Explain the concept of torsionally equivalent shaft and derive equation for its equivalent length. [7]
- b)** Determine the natural frequency and corresponding mode shapes of given system shown in Fig. 5b when $K = 40\text{N/m}$ and $m = 10\text{kg}$. [10]

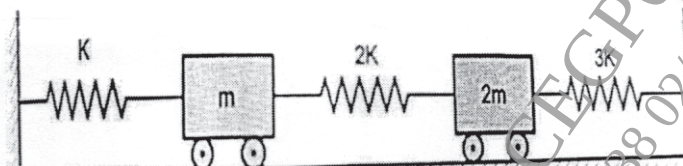


Fig. 5b

OR

- Q6) a)** Explain the torsional vibrations of geared system neglecting inertia of gears. [7]
- b)** For a spring-mass system shown in Fig. 6b, find Eigen values and Eigen vectors by matrix method. [10]

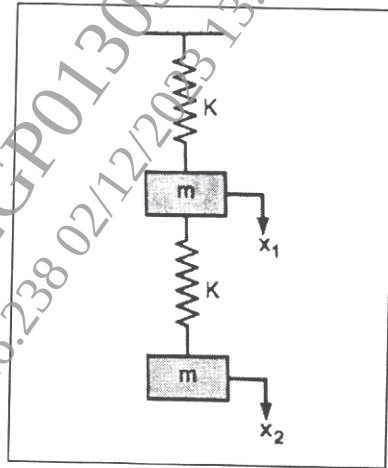


Fig. 6b

- Q7) a)** What is FFT analyzer? Explain the importance of frequency spectrum for machinery fault detection. [10]
- b)** Explain the process of vibration measurement with neat sketch. [8]

OR

- Q8) a)** What are different methods of vibration control? Explain any one in detail. [6]
- b)** Explain with neat sketch the working of sound level meter. [6]
- c)** Explain anechoic chamber and reverberant chamber. [6]

